

Data Mining

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1 Introduction

Data mining, the collection and processing of large sets of data, has become the new way of predicting the future.

2 History

Despite having been around for hundreds of years, data mining (originally termed knowledge discovery in databases, or KDD) was officially introduced by Gregory Piatetsky Shapiro in 1989. This method was intended to grow and increase profitable production in companies through the identification of trends from provided data. Data mining combines the works of various statisticians across several different time periods, with the following three being crucial to the process.

2.1 Bayes's Theorem (1700s)

$$P(A | B) = \frac{P(B | A) \cdot P(A)}{P(B)}$$

Named after mathematician Thomas Bayes, this formula states that the likelihood of an event occurring (event A) given that another event has already occurred/is true (event B), $P(A | B)$, can be calculated by multiplying the probability of event A occurring, $P(A)$, by the chances of event B occurring given that event A is true, $P(B | A)$, and dividing the product by the probability of event B occurring. This formula can predict the chances of an event happening while taking other factors into consideration. To put this in a consumer-seller perspective, Bayes' theorem can answer questions such as, how likely are customers to impulsively purchase snacks from row 7 of the store after they have finished shopping for gadgets in row 5? Apart from determining marketing and financial strategies, Bayes' theorem can also be applied to the engineering field, medical field, sports and the legal profession. Due to its ability to detect the relationship between an event and complex data sets relative to that event, this theorem is incorporated into almost all forms of data mining.

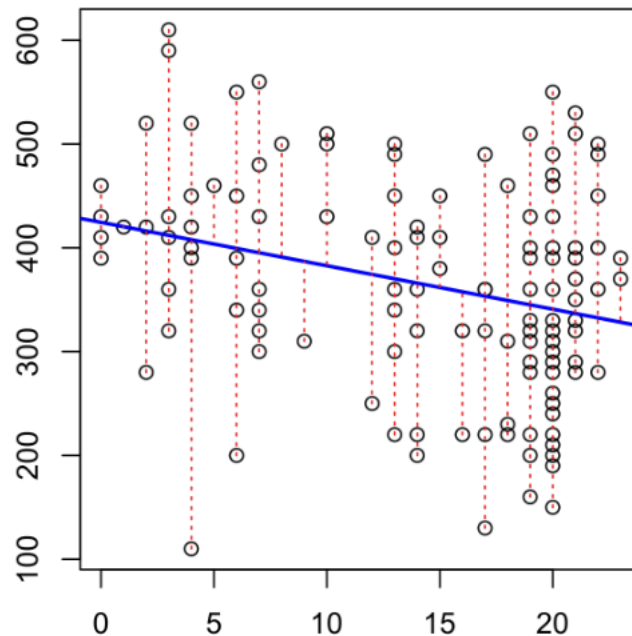
2.2 Regression Analysis (1800s)

$$A = \frac{(\Sigma y)(\Sigma x^2) - (\Sigma x)(\Sigma xy)}{n(\Sigma x^2) - (\Sigma x)^2}$$
$$B = \frac{n(\Sigma xy) - (\Sigma x)(\Sigma y)}{n(\Sigma x^2) - (\Sigma x)^2}$$

In 1805, Adrien-Marie Legendre and Johann Carl Friedrich Gauss published the method that is the foundation for regression analysis: least squares. One type of regression analysis is linear regression, which describes a universally observed phenomenon where the measured values of an event

are much more likely to experience a decrease/regression towards the mean following an immediate increase in values. The basic formula of linear regression creates a best fitting line for a series of plotted data points on a graph to predict future trends, and it can be simplified down to the slope intercept formula, $y = a + bx$, with variables b (B) and a (A) broken down into more complex least squares equations. To calculate the slope of the best fitting line, the product of the sum of all x values (Σx) and the sum of all y values (Σy) must be found. Then, the sum of the products between the x -value and the y -value of each data point (Σxy) has to be multiplied with the total number of plotted points (n). Subtracting the two products will give us the expression $n(\Sigma xy) - (\Sigma x)(\Sigma y)$, or the numerator of B . To find the denominator of B , all x values must be squared and added together (Σx^2) before being subtracted from the product of n (the amount of plotted points) and itself: $n(\Sigma x^2) - (\Sigma x)^2$. To find B , divide these two differences. Using the same variables as the formula for slope, the y -intercept of this equation (A) is found by multiplying the sum of all x -values and the sum of all y -values (Σxy). That product is then multiplied by the sum of all x values to form the expression $(\Sigma x)(\Sigma xy)$. Finally, the sum of all squared x values (Σx^2) is multiplied by the sum of all y values (Σy). The difference between the two products gives us $(\Sigma y)(\Sigma x^2) - (\Sigma x)(\Sigma xy)$. This is then placed over the same denominator as B , $n(\Sigma x^2) - (\Sigma x)^2$.

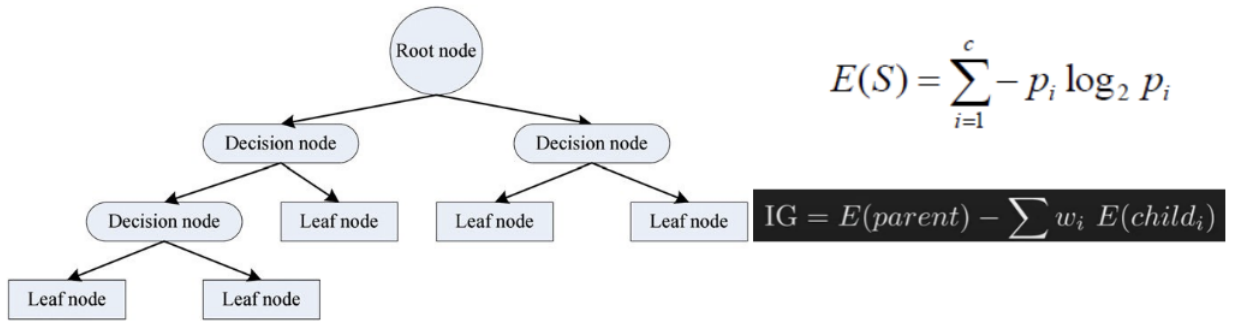
The entire completed equation for linear regression is $y = a + bx + \epsilon$, where the greek letter epsilon represents the deviation of a data point from the best fit line (as shown in the graph below by red dotted lines). To find ϵ of each point, the expected y value must be subtracted from the actual y value. When all values of ϵ are added up, the sum should equal to zero.



2.3 Decision Trees (1900s)

The exact roots of decision trees are unknown with multiple researchers contributing to its development around the mid-1900s. The tree starts off with the root node, where all the unsorted information is stored before branching off into either leaf nodes or more decision nodes. Leaf nodes hold the final output, or information that cannot be sorted further and are already in a set. The

nodes are connected by branches, which sorts the data from a previous node given a set of conditions. For the system to categorize information the most optimal way, it calculates entropy and information gain at an extremely fast pace. Entropy, in this case, is the measure of certainty of a randomly chosen point from a node and its predicted class corresponding to each other. To calculate the entropy inside a node, find the probability of choosing a data point belonging to a certain class within all the data points in that specific node, and multiply that number by $\log_2$ of itself. Do so with all the classes of points, and find the difference between all of them. An entropy of zero guarantees absolute certainty of accurate correspondence and also indicates a pure node, meaning all data inside that node belongs to a singular class. Since each split produces two nodes, the information gain must be calculated by subtracting the combined entropy of the child nodes from the entropy of the parent node. The entropy of a child node relative to the parent is found through multiplying the previously obtained entropy value by the number of data points in the child node out of the total number of data points in the parent node. After this is calculated for both child nodes, the values are subtracted from the entropy of the parent node. The formation of the split with the greatest value for information gain is then chosen by the system to continue the decision tree.



3 Techniques

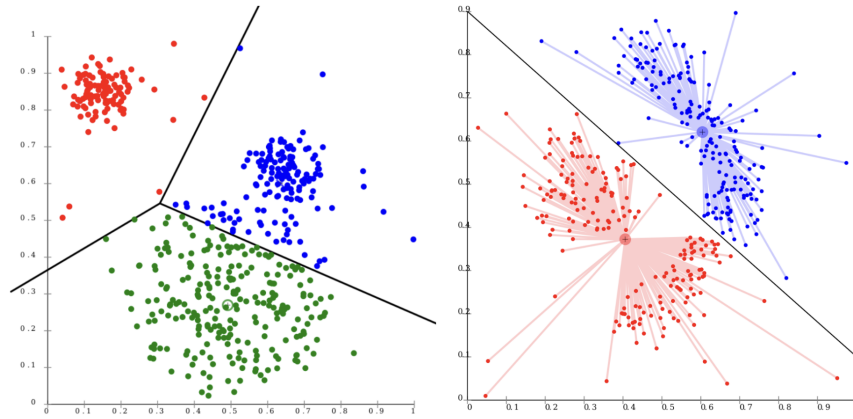
After accumulating mass amounts of data from sources such as a data warehouse or a data lake, a combination of classified and unclassified data must be sorted. Many apps or websites use APIs to collect structured data from various sources that contain data stored in the form of files. As for unstructured data, techniques listed below are required to clean and organize all of the collected information.

3.1 Clustering

A common method used to organize collected data into data sets is clustering, which is an umbrella term for various types of algorithms. Generally, sets of data are separated into classes based on their similarities and data types. Different kinds of data require different algorithms to perform clustering, for example, unstructured data requires algorithms that are nonsensitive to missing or false data. Some popular clustering methods are as follows:

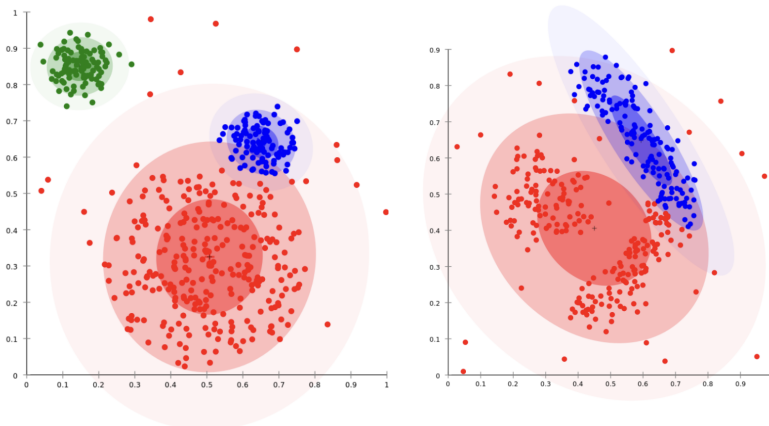
3.1.1 Partitioning Method

In this method, the data is grouped into clusters so that the size of each individual cluster is less than or equal to the total data items ($k \leq n$). The two rules that must be satisfied for partitioning to work are: each cluster should contain at least one object, and objects cannot belong to more than one cluster. The classification of data uses the Euclidean distance formula, $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$, which measures the distance between two points, specifically data points in this case. Using the first two data points, two clusters are formed and labeled as different centroids. The distance of the third data point from the two different centroids is then measured, and then grouped with the centroid/cluster it is closer to. Finally, the centroid of the cluster with the additional object (third data point) is redetermined by finding the midpoint between the two x values and y values. This pattern continues until all the data points are partitioned.



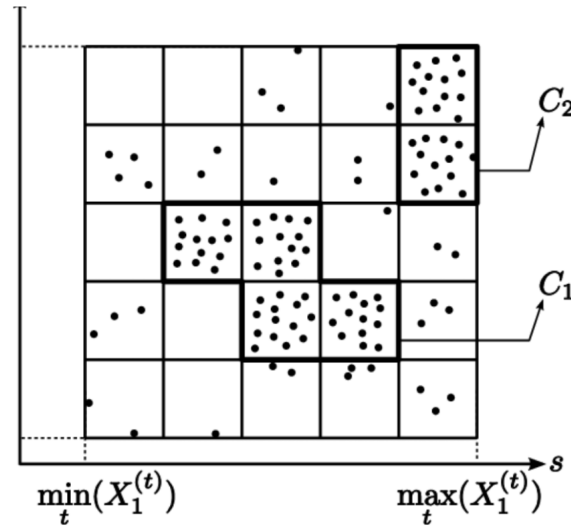
3.1.2 Density-Based Method

The principal idea is that data objects are grouped based on density (mass/volume), with the goal of identifying regions that have high density. After points are plotted on a graph, a point (P) is chosen along with a random real number (ϵ). A circle known as a neighborhood with radius equal to ϵ is formed around point P that allows the density of that specific neighborhood to be calculated. Mass is represented by the amount of points in that neighborhood, which is divided by the area of the circle around point P , taking the place of volume. This process is repeated with various points until all clusters have been identified and all points have been grouped. Outliers are determined by neighborhoods with extremely low density and labeled as “noise” and disregarded.



3.1.3 Grid-Based Method

Grid-Based clustering is faster than the two previous methods, and can identify outliers unlike the partitioning method. First, the provided data is plotted and divided into “cells” with the same width intervals. In the specific STING (Statistical Information Grid) clustering technique, the data is split into layers and divided into increasing numbers of cells going down each layer. Information such as confidence interval or estimated range of probability is calculated for each cell to determine the relevance of that cell, which then helps identify clusters of data. The overall idea of this process is to check for the number of data points inside each cell as well as its surrounding cells. If the statistical information of the cells meet set requirements, neighboring cells are grouped together as long as the threshold (minimum amount of mass/points per cell) is exceeded.



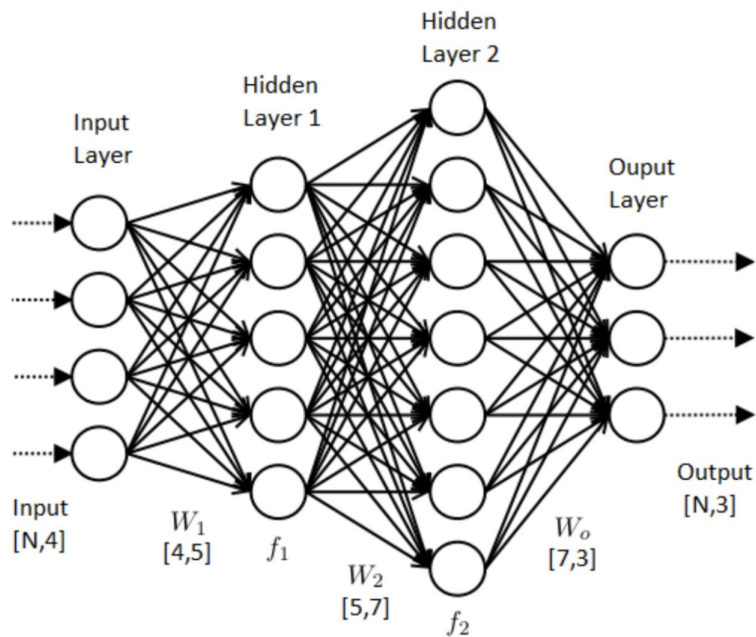
3.2 Machine Learning

This type of algorithm involves training computer programs or software to create models based on preexisting sample data to instill a passive pattern-predicting behavior in the machine. Data mining primarily focuses on unsupervised learning, which generates new content rather than analyze and recognize patterns in provided data sets. Unsupervised learning generates information from mimicking uncategorized data it finds and self-categorizes it based on random probability densities.

3.2.1 Neural Networks

Neural networks utilizes mathematical concepts and processing on the basis of biological means to adapt its structure based on the incoming flow of information it receives. The model of a neural network consists of artificial neurons and its adaptations are thought to be similar to a human brain; The algorithm adjusts according to the Hebbian theory, which states that repeated and persistent stimulation of a postsynaptic cell causes an increase in the effectiveness of synaptic transmissions. This type of conditioning is done so that a trend can be identified in a data set without any prior exposure to that specific class of data. Neural networks consist of many layers of neurons, also known as the processing units of the network. The input layer receives information and the output layer makes predictions. These two layers surround the hidden layers, where most of the processing occurs. Each pixel of an image is fed through one neuron of the input layer, which is connected to

other layers by channels that each hold a numerical value between, or weight. To find the weighted sum, multiply the weight of the channel by the numerical value assigned to the input, in this case, how bright/dark the pixel is. During this process, there is bound to be some margin of error. To compensate for that error, a numerical bias associated with each node is added to the sum, and then passed through an activation function. The activation function acts as a threshold. For example, if the sum is equal to or greater than a set value, then a signal will be sent to certain nodes in the second hidden layer. This is repeated as many times as needed until the output layer is reached, which then produces an output from the node with the greatest value. To train the software, that output is compared with the desired/correct output, and adjustments are made to reduce errors.



3.3 Prediction

This ability enables softwares to produce models that project future behavior given a large set of existing data. Some examples of prediction are forecast modeling, classification modeling and time series modeling. Forecast modeling is often used to predict demand of a product, consumer behavior and overall sale outcomes. It involves anything from predicting constant growth rate with minimum mathematical calculations to multiple linear regression, which accounts for more than one independent variable. Classification modeling predicts the categorization of future incoming data based on pre-existing trends. The program looks for recurring patterns in historical data that might distinguish between different types of received data. An example of this is categorizing spam emails. Time series modeling pinpoints a specific point or period of time in the future, and is more commonly applied in cryptocurrency trends, financial markets, sports analytics, and weather forecasting. This type of modeling can also graph individual predictions dependent on trend, cycle, seasonality and remainder, which is known as decomposition.

4 Conclusion

All in all, data mining is a process that can better our understanding of the world. Apart from growing businesses and recognizing trends in events, the practices within data mining improves the development of artificial intelligence such as machine learning. Although the predictions of the future using these tools are not completely reliable as of right now, we are slowly inching towards that reality.

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